Hasan Kurban

· Trustworthy AI & Foundation Model Interpretability · Reasoning & Inference-Time Compute
· Agentic AI · Multimodal Foundation Models · AI for Science

Ph.D. in Computer Science, Indiana University Bloomington, 2017

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159 Papers • **81** Journal Articles • **78** Conference Papers • **3** Book Reviews

16 Graduate Researchers • **28** Undergraduate Researchers

17 Funded Grants • **$8.7M** Research Funding • *h*-index **16**

Biographical Sketch. Hasan Kurban is an Assistant Professor at Hamad Bin Khalifa University and Founding Director of the *Kurban Intelligence Lab*, with adjunct appointments at Indiana University Bloomington and Texas A&M University at Qatar. His program is organized around a measurement problem: as models outrun human verification, what can be established about their behavior, and with what guarantees? Frontier systems are now graded largely by other models, and the lab asks what that substitution identifies and what it silently loses. The technical core is **identifiability**, the question of which quantities an evaluation protocol recovers and which it merely appears to. Recent results establish that multilingual LLM-judge rankings reverse under a bias a label-free double-centering estimator removes, that models are routinely correct while ungrounded in the evidence they cite, and that agent-as-a-judge verdicts track prompt language and backbone rather than requirement satisfaction. No negative result is published without the corrected estimator, blind baseline, or distribution-free certificate that repairs it.

The same question recurs across the lab's other lines. In **reasoning and inference-time compute**, whether longer chains buy accuracy or only its appearance. In **interpretability**, whether sparse autoencoders localize hallucination precisely enough to suppress it. In **agentic systems**, how errors compound across turns and languages, and what a sound audit costs. In **multimodal learning**, whether world models recover physical dynamics or only pixel statistics. In **scientific machine learning**, whether a learned prediction survives physical verification: Clifford-algebra equivariant interatomic potentials, symmetry constraints that reject physically impossible outputs, and statistically valid screening under a limited first-principles budget. The work extends to materials science and quantum chemistry, biomedical imaging, energy systems, and genomics with domain scientists as co-authors, and every paper is released with its benchmarks, datasets, and code through the lab's GitHub organization.

The group's work has appeared at *NeurIPS*, *ICML*, *ICLR*, *CVPR*, *ECCV*, *ICCV*, *ACL*, *EMNLP*, *COLM*, *IJCAI*, and *KDD*, and in Nature Portfolio journals, *IEEE Transactions*, and *Machine Learning*. He is an editor for Nature Portfolio, organizes an ArabicNLP shared task, and holds sixteen competitive grants including QRDI national programs on agentic AI governance. Alumni are at Meta, ServiceNow, Eli Lilly, and the University of Maryland; his undergraduates have first-authored at *NeurIPS*.

Contents

Research Interests

Each area is stated as the question that drives it.

- **Trustworthy AI and Foundation Model Interpretability.** *What does an evaluation protocol actually identify, and when can a model's output be certified?* Identifiability and corrected estimators for paired and multilingual LLM evaluation; hallucination detection and mitigation; sparse-autoencoder feature analysis and model editing; calibration, selective prediction, and abstention; jailbreak robustness and red-teaming.
- **Reasoning and Inference-Time Compute.** *When does additional test-time computation buy accuracy rather than the appearance of it?* Reinforcement learning with verifiable rewards (RLVR);

posterior-constrained inference for mathematical reasoning; decoding-format confounds in consistency-based selection; multi-agent and neurosymbolic reasoning; robustness of chain-of-thought under noisy, conflicting, and adversarial evidence.

- **Agentic AI and LLM Agents.** *How do errors compound over long horizons, and what does a sound audit cost?* Tool use and long-horizon planning; retrieval-augmented and knowledge-grounded reasoning; agent-as-a-judge evaluation and its language and backbone sensitivity; cost-aware audit protocols; agent safety; scientific and biomedical discovery agents.
- **Multimodal Foundation Models.** *Do multimodal systems recover the structure of the world, or only correlations in the training distribution?* Vision–language models; video understanding and long-context temporal reasoning; world models and inverse identification of physical dynamics from video; conformal coverage for temporal grounding; embodied and spatial reasoning.
- **AI for Science.** *Can a learned model make a scientific claim that survives physical verification?* Equivariant and symmetry-constrained interatomic potentials; physics-informed learning; statistically valid screening under a limited first-principles budget; multimodal benchmarks and active learning for materials informatics, quantum chemistry, biomedical imaging, and energy systems.

Research Funding and Grants

Competitive Research Grants (PI / Lead PI)

2026 **QRDI: Graduate Student Research Award (GSRA).** Efficient Test-Time Reasoning in Compressed Vision–Language Models.

Role: Faculty Supervisor • Student: Mr. Marcus Wein Monteiro • Award No. GSRA13-L-0423-260214 • Budget: $46,093.48.

2026 **QRDI: Agentic Artificial Intelligence Grant.** A Governance and Audit Toolkit for the Responsible Adoption of Agentic Artificial Intelligence in Qatar's Financial Sector.

Role: Lead Principal Investigator • Award No. AAIC01-0130-260007 • Budget: $41,000.

2026 **QRDI: Agentic Artificial Intelligence Grant.** Agentic AI Assurance for Qatar: Global Governance Mapping, GCC Benchmarking, and a Finance-Focused Policy Toolkit.

Role: Principal Investigator • Award No. AAIC01-0208-260013 • Budget: $40,978.

2025 **QRDI: Modernizing Charitable Sector Grant.** Advancing AI-Driven Governance in Qatar's Charitable Sector: A Trustworthy Arabic LLM for Regulatory Compliance.

Role: Lead Principal Investigator • Award No. MCSC02-0221-250019 • Budget: $40,999.

2025 **QRDI: Undergraduate Research Experience Program (UREP).** Leveraging Data Analytics for Interpreting Transformer Dissolved Gas Analysis.

Role: Principal Investigator • Award No. UREP31-043-2-014 • Budget: $26,042.

2025 **TÜBİTAK 1001 Scientific and Technological Research Projects Support Program.** Improving Dynamic Fit and Comfort of Prosthetic Sockets Using Wearable Fiber Optic Sensors.

Role: Principal Investigator • Award No. 125E223 • Budget: $60,000.

2023 **QRDI: National Priorities Research Program (NPRP).** ReSolVE: Resilient Solutions for Vulnerabilities and Emergencies: An Effective National Risk Management Plan for Qatar.

Role: Principal Investigator (joined 2025) • Award No. NPRP14C-0909-210008 • Budget: $4,434,601.

2021 **QRDI: National Priorities Research Program Cluster (NPRP-C).** Multi-layer Cyber-security and Situational Awareness to Enhance Resiliency in Qatar's Power Grid.

Role: Principal Investigator (joined 2026) • Award No. NPRP12C-0814-190012 • Budget: $3,204,904.

Institutional Grants and Internal Funding

2026 **Student Research Experience Grant (SREG),** Texas A&M University at Qatar. Physics-Guided 4D Gaussian Splatting for Sparse-View Dynamic Scene Understanding.

Role: Principal Investigator • Budget: $44,000.

2025 **HBKU Horizon Hub 1st Cycle Program,** Office of the VP for Research. AI for Science Symposium (January 21–22, 2026).

Role: Principal Investigator • Award No. HBKU-INT-VPR-HHW-01-8 • Budget: QAR 150,000.

2025 **HBKU Thematic Research Grant Program (3rd Cycle).** Optimal AI and Quantum-Secured Wireless–Optical Communication Design for Smart Cities in Qatar.

Role: Principal Investigator • Award No. HBKU-INT-VPR-TG-03-04 • Budget: $102,740.

2025 **HBKU GPU Compute Grant,** Research Computing Core Group. Compressed Motion Statistics for Reliable Evaluation of Generative Videos.

Role: Lead Principal Investigator • Budget: QAR 7,000.

2024 **Transformative Educational Experience (TEE) Grant,** Texas A&M University at Qatar. Transforming Software Engineering Education through Autograder and LLM Integration.

Role: Lead Principal Investigator • Budget: $10,000.

2024 **Multiversity Academic Grant,** Texas A&M University at Qatar. A Novel, Multi-Institutional Machine Learning for Engineers Course.

Role: Lead Principal Investigator • Budget: $10,000.

2023 **Faculty Resource Allocation (Three-Year),** Texas A&M University at Qatar.

Role: Lead Principal Investigator • Budget: $53,750.

2023 **Research Impact Initiative,** Texas A&M University at Qatar. Leveraging AI, Machine Learning, and Data Analytics to Enhance Qatar's Sustainable Energy, Healthcare, and Security.

Role: Lead Principal Investigator • Budget: $177,000.

2023 **Startup Research Funds (Three-Year),** Texas A&M University at Qatar.

Role: Lead Principal Investigator • Budget: $413,000.

Selected Publications

18 representative works, drawn from the 162 papers listed in full below. Journal entries are ordered by venue impact factor (Clarivate JCR, most recent release); conference entries by venue standing. **Bold** denotes Dr. Kurban; a dagger (†) denotes corresponding author. Workshop and special-track papers are labeled with their track.

Award and Distinction

S1 **Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding.**
C. Polat, **H. Kurban**†, E. Serpedin, M. Kurban.
42nd IEEE/CVF Computer Vision and Pattern Recognition Conf. (CVPR, MM4Mat), Nashville, TN, USA, 2025. • Spotlight

Flagship AI Conferences

S2 **Perception, Not Reasoning, Limits Visual Theory of Mind.**
M. R. Barhdadi, S. T. Wasim, J. Gall, E. Serpedin, **H. Kurban**†.
40th Conf. on Neural Information Processing Systems (NeurIPS), Sydney, Australia, 2026.

S3 **IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video.**
R. Khanbayov, M. R. Barhdadi, E. Serpedin, **H. Kurban**†.
19th European Conf. on Computer Vision (ECCV), Malmö, Sweden, 2026.

S4 **How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science.**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†.
32nd ACM SIGKDD Conf. on Knowledge Discovery and Data Mining (KDD), Jeju, Korea, 2026.

S5 **Correct but Not Grounded: Stress Testing Evidence Utilization in Large Language Models.**
S. Abdaljalil, E. Serpedin, **H. Kurban**†.
31st Conf. on Empirical Methods in Natural Language Processing (EMNLP), Budapest, Hungary, 2026.

S6 **Rank Reversal in Multilingual LLM Judges: A Label-Free Double-Centering Calibrator.**
A. S. Mahmood, S. Abdaljalil, **H. Kurban**†.
31st Conf. on Empirical Methods in Natural Language Processing (EMNLP), Budapest, Hungary, 2026.

S7 **Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation.**
A. S. Mahmood, S. Abdaljalil, **H. Kurban**†.
3rd Conf. on Language Modeling (COLM), San Francisco, CA, USA, 2026.

S8 **EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration.**
M. R. Barhdadi, M. Tuncel, E. Serpedin, **H. Kurban**†.
39th Conf. on Neural Information Processing Systems (NeurIPS), San Diego, CA, USA, 2025.

S9 **SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs.**
S. Abdaljalil, F. Pallucchini, A. Seveso, **H. Kurban**†, F. Mercorio, E. Serpedin.
30th Conf. on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China, 2025.

High-Impact Journals

S10 **Agentic AI in Biomedical and Healthcare: A Modality-Centric Review of Emerging Frameworks, Open Challenges, and Future Directions.**
M. R. Islam, M. M. Islam, M. Y. Ansari, **H. Kurban**†, E. Serpedin.
Artificial Intelligence Review, 2026.

S11 **CliffordIP: Clifford Algebra Equivariant Interatomic Potentials for Heterogeneous Catalysis.**
C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†.
npj Computational Materials (Nature Portfolio), 2026.

S12 **Linear-Complexity Unified Defense Against Deception Attacks in Distributed Economic Dispatch Using Cryptography and Machine Learning.**
M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin.
IEEE Transactions on Smart Grid, 2026.

S13 **Multimodal Explainable AI-Driven Analysis of Quantum Size Effects in Copper Nanoclusters for Hydrogen Storage.**
M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†.
International Journal of Hydrogen Energy, 2026.

S14 **Bayesian Probabilistic Knowledge from Diameter Prior for Decision Fusion to Detect Lung Nodule Heterogeneity.**
M. R. Islam, M. K. Hasan, **H. Kurban**†, E. Serpedin.
IEEE Transactions on Artificial Intelligence, 2026.

S15 **Geometric-*k*-means: A Bound-Free Approach to Fast and Eco-Friendly *k*-means.**
P. Sharma, M. S. Malec, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic.
Machine Learning, 2025.

Track and Workshop Papers at Flagship AI Venues

S16 **Data Expressiveness and Its Use in Data-centric AI.**
H. Kurban, P. Sharma, M. M. Dalkilic.
35th Conf. on Neural Information Processing Systems (NeurIPS, Data-Centric AI), 2021.

S17 **xChemAgents: Agentic AI for Explainable Quantum Chemistry.**
C. Polat, M. Tuncel, M. Kurban, E. Serpedin, **H. Kurban**†.
42nd Int. Conf. on Machine Learning (ICML, Multi-Agent Systems in the Era of Foundation Models), Vancouver, Canada, 2025.

S18 **Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models.**
S. Abdaljalil, **H. Kurban**†, K. Qaraqe, E. Serpedin.
63rd Annual Meeting of the Association for Computational Linguistics (ACL, Towards Knowledgeable Foundation Models), Vienna, Austria, 2025.

Publications

Peer-Reviewed Journal Articles

Items marked *(under review)* at the right margin are currently under peer review.

2026

[81] M. R. D. Meeran, S. Abdaljalil, **H. Kurban**†. "Auditing Chain-of-Thought Faithfulness Scores for LLM Safety Monitoring," *IEEE Transactions on Dependable and Secure Computing*, 2026. UNDER REVIEW

[80] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. "When do machine-learned exchange–correlation improvements inherit into density-functional tight binding?."*npj Computational Materials*, 2026. UNDER REVIEW

[79] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. "A single design choice determines whether machine learning models of materials make physically impossible predictions."*Nature Communications*, 2026. UNDER REVIEW

[78] **H. Kurban**, R. Khanbayov, M. Kurban. "Physics as the label for measuring and correcting materials reasoning in multimodal models." *npj Computational Materials*, 2026. UNDER REVIEW

[77] V. Tataroglu, M. O. Cakiroglu, G. Gunduz, H. Bulut, M. M. Dalkilic, **H. Kurban**†. "Protocol Before Architecture: Auditing Forecasters Against the Random Walk, Evidence from 46 Systems on Daily Bitcoin." *Expert Systems with Applications*, 2026. UNDER REVIEW

[76] I. B. Altun, E. K. Buxton, M. M. Dalkilic, **H. Kurban**†. "The Shape of AI Safety: A Cartographic Synthesis and Analysis of 12,987 Papers from Premier Venues (2018–2026)." *Artificial Intelligence Review*, 2026. UNDER REVIEW

[75] V. Tataroglu, **H. Kurban**†. "Pricing Judge Dependence in Staked LLM Committees." *Information Sciences*, 2026. UNDER REVIEW

[74] M. A. Yatkin, M. Kõrgesaar, **H. Kurban**. "Embedding Consistent Recurrent Neural Networks in Finite Element Simulations for Path-Dependent Damage Prediction," *Applied Sciences*, 2026. UNDER REVIEW

[73] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. "Grounded verification of chemical and materials reasoning: detection is the bottleneck."*Communications AI & Computing*, 2026. UNDER REVIEW

[72] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Local motif-guided discovery of hydrogen-stabilizing environments in refractory high-entropy alloys," *Journal of Alloys and Compounds*, 2026. UNDER REVIEW

[71] **H. Kurban**, M. Kurban. "Certified discovery: statistically valid materials screening claims from a small DFT verification budget. " *Digital Discovery*, 2026. UNDER REVIEW

[70] **H. Kurban**, M. Kurban. "Materials foundation models fail together: correlated errors and collective blind spots. " *Computational Materials Science*, 2026. UNDER REVIEW

[69] V. Tataroglu, **H. Kurban**†. "The Persuasion–Prediction Gap: Outcome-Grounded Auditing of LLM Judges on Market-Resolved Financial and Crypto Reasoning." *Expert Systems with Applications*, 2026. UNDER REVIEW

[68] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Site-engineered reversible H_2 adsorption on Zr-doped anatase TiO_2 nanoparticles: active learning-assisted hydrogen storage screening." *International Journal of Hydrogen Energy*, 2026. UNDER REVIEW

[67] S. Abdaljalil, E. Serpedin, K. Qaraqe, **H. Kurban**†. "Audit-of-Understanding: Posterior-Constrained Inference for Mathematical Reasoning in Language Models," *IEEE Access*, 2026. UNDER REVIEW

[66] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine-Robust, Privacy-Preserving Distributed Unit Commitment Using a Hybrid Approach," *IEEE Transactions on Industrial Cyber-Physical Systems*, 2026. UNDER REVIEW

[65] M. M. Islam, M. Ismail, A. Takiddin, **H. Kurban**, R. Atat, E. Serpedin. "Evaluating False Data Injection Detectors in Smart Grid Under Stealthy, High-Impact Attacks," *IEEE Transactions on Smart Grid*, 2026. UNDER REVIEW

[64] M. R. Islam, S. Abdaljalil, E. Serpedin, **H. Kurban**†. "ConfTriage: Calibration-Aware Large Language Model Triage for Pulmonary Nodule Malignancy with Selective Specialist Deferral," *IEEE Journal of Biomedical and Health Informatics*, 2026. UNDER REVIEW

[63] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. "SCALAR: Quantifying Structural Hallucination, Consistency, and Reasoning Gaps in Materials Foundation Models."*Digital Discovery*, 2026. UNDER REVIEW

[62] C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. "QuantumCanvas: A Multimodal Benchmark for Learning Two-Body Quantum Interactions."*Machine Learning: Science and Technology*, 2026. UNDER REVIEW

[61] R. Khanbayov, **H. Kurban**†. "Topical Phase Transitions in Artificial Intelligence Research: Large-Scale Evidence and an Early-Warning Signature for Emerging Topics," *Quantitative Science Studies*, 2026. UNDER REVIEW

[60] M. Elnour, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. "Robust Graph Contrastive Learning for Attack Detection in Power Distribution Systems," *IEEE Open Access Journal of Power and Energy*, 2026. UNDER REVIEW

[59] M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. "Enhancing Robustness of FDI Attack Detectors in Smart Grids Against Adaptive Adversaries via a Physics-Constrained GAN," *Electric Power Systems Research*, 2026. UNDER REVIEW

[58] M. M. Islam, M. Ismail, **H. Kurban**, A. Takiddin, E. Serpedin. "Byzantine-Resilient Distributed DC Optimal Power Flow Using Cryptography and Machine Learning," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. —— under review

[57] M. Kurban, E. Serpedin, **H. Kurban**. "Sequential Molecular Hydrogen Loading on the ZIF-8 Metal–Organic Framework: Adsorption Hierarchy, Electronic Structure, and Charge Transfer Mechanism," *International Journal of Hydrogen Energy*, 2026. —— under review

[56] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Physics-Guided Thermodynamic Screening of H_2 Adsorption on Alkaline-Earth-Modified Anatase TiO_2 Nanoparticles with LLM-Assisted Interpretation ," *RSC Advances*, 2026. —— under review

[55] M. Elnour, M. AlShaikh Saleh, M. O. Cakiroglu, **H. Kurban**, R. Atat, A. Takiddin, M. Ismail, K. R. Davis, E. Serpedin. "A Deep-Learning Based Approach for Photovoltaic Forecasting Using Symbolic Graph Representations," *IEEE Open Access Journal of Power and Energy*, 2026. —— under review

[54] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine Fault-Tolerant Lightweight Security Framework for Distributed DC Optimal Power Flow Using Threshold Cryptography," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. —— under review

[53] S. Abdaljalil, E. Serpedin, **H. Kurban**†. "HalluVerse3: A Fine-Grained Multilingual Benchmark for Hallucination Detection in LLMs," *Language Resources and Evaluation*, 2026. —— under review

[52] M. M. Islam, M. Ismail, **H. Kurban**, E. Serpedin. "Byzantine-Resilient, Privacy-Preserving, Scalable Distributed Unit Commitment Using Threshold Cryptography," *IEEE Open Journal of Power Systems Society*, 2026. —— under review

[51] M. Elnour, M. AlShaikh Saleh, R. Atat, X. Huo, A. Takiddin, M. Ismail, **H. Kurban**, K. R. Davis, E. Serpedin. "Joint Sensor Deployment and Physics-Informed Graph Transformer for Smart Grid Attack Detection," *IEEE Open Access Journal of Power and Energy*, 2026. —— under review

[50] M. R. Islam, M. Y. Ansari, **H. Kurban**†, E. Serpedin. "A Multimodal Interpretable Approach to Lung Nodule Diagnosis from CT with Radiology and Anatomy-Aware Text," *IEEE Transactions on Radiation and Plasma Medical Sciences*, 2026. —— under review

[49] I. B. Altun, M. O. Cakiroglu, S. Awadallah, M. M. Dalkilic, **H. Kurban**†. "Benchmarking Artificial Intelligence Models for Dissolved Gas Forecasting in Power Transformers," *Engineering Applications of Artificial Intelligence*, 2026. —— under review

[48] C. Polat, M. Kurban, E. Serpedin, **H. Kurban**†. "CliffordIP: Clifford Algebra Equivariant Interatomic Potentials for Heterogeneous Catalysis," *npj Computational Materials*, 2026.

[47] M. R. Islam, M. M. Islam, M. Y. Ansari, **H. Kurban**†, E. Serpedin. "Agentic AI in Biomedical and Healthcare: A Modality-Centric Review of Emerging Frameworks, Open Challenges, and Future Directions," *Artificial Intelligence Review*, 2026.

[46] M. Kurban, E. Serpedin, **H. Kurban**. "Hydrogen Storage in TiFe Nanoparticles: Void-Limited Saturation and Metal-Selective Charge Transfer During Sequential Loading," *International Journal of Hydrogen Energy*, 2026.

[45] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Explainable Machine Learning Reveals Orbital Design Rules for Reversible H_2 Adsorption on Single-Atom-Doped TiO_2 Nanoparticles," *Materials & Design*, 2026.

[44] M. M. Islam, M. Ismail, **H. Kurban**, X. Huo, E. Serpedin. "Lightweight Defense Against Data Consistency Attacks in Distributed DC Optimal Power Flow," *IEEE Transactions on Automation Science and Engineering*, 2026.

[43] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Descriptor-guided Thermodynamic Screening of H_2 Adsorption on Single-atom-doped Anatase TiO_2 Nanoparticles with Interpretable Machine Learning," *Scientific Reports*, 2026.

[42] M. M. Islam, A. Takiddin, M. Ismail, **H. Kurban**, E. Serpedin. "Linear-Complexity Unified Defense Against Deception Attacks in Distributed Economic Dispatch Using Cryptography and Machine Learning," *IEEE Transactions on Smart Grid*, 2026.

[41] M. A. Yatkin, M. Kõrgesaar, V. M. Asan, J. Romanoff, J. Stuckner, **H. Kurban**. "Self-Consistent Recurrent Neural Network for Path Dependent Deformation," *Scientific Reports*, 2026.

[40] S. Abdaljalil, P. Sharma, R. Atat, E. Serpedin, **H. Kurban**†. "SINdex: Semantic INconsistency Index for Hallucination Detection in LLMs," *IEEE Open Journal of the Computer Society*, 2026.

[39] M. R. Islam, M. K. Hasan, **H. Kurban**†, E. Serpedin. "Bayesian Probabilistic Knowledge from Diameter Prior for Decision Fusion to Detect Lung Nodule Heterogeneity," *IEEE Transactions on Artificial Intelligence*, 2026.

[38] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Multimodal Explainable AI-Driven Analysis of Quantum Size Effects in Copper Nanoclusters for Hydrogen Storage," *International Journal of Hydrogen Energy*, 2026.

[37] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Engineered MgO Nanoparticles with Tunable Electronic Signatures for Energy Applications," *ACS Applied Nano Materials*, 2026.

2025

[36] H. Cetinkaya, F. Ay, M. Tuncel, H. Nounou, M. N. Nounou, **H. Kurban**, E. Serpedin. "Curriculum-Enhanced Adaptive Sampling for Physics-Informed Neural Networks: A Robust Framework for Stiff PDEs," *Mathematics*, 2025.

[35] **H. Kurban**, P. Sharma, M. M. Dalkilic, M. Kurban. "Accelerating Density of States Prediction in Zn-Doped MgO Nanoparticles via Kernel-Optimized Weighted k-NN," *Scientific Reports*, 2025.

[34] S. Abdaljalil, **H. Kurban**†, R. Atat, E. Serpedin, K. Qaraqe. "Deep Temporal and Structural Embeddings for Robust Unsupervised Anomaly Detection in Dynamic Graphs," *IEEE Open Journal of the Computer Society*, 2025.

[33] M. K. Ramachandra, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic. "Telescope Indexing for k-Nearest Neighbor Search Algorithms over High-Dimensional Data & Large Data Sets," *Scientific Reports*, 2025.

[32] P. Sharma, M. S. Malec, **H. Kurban**†, M. O. Kulekci, M. M. Dalkilic. "Geometric-k-means: A Bound-Free Approach to Fast and Eco-Friendly k-means," *Machine Learning*, 2025.

[31] M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**†, M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin. "An Extended Frequency-Improved Legendre Memory Model for Enhanced Long-term Electricity Load Forecasting," *IEEE Open Access Journal of Power and Energy*, 2025.

[30] C. Polat, **H. Kurban**†, M. Kurban. "Enabling Ease of Access to Quantum Chemistry with Transformer-Based Text Encoding and Physics-Informed Multilayer Perceptron," *Physica Scripta*, 2025.

[29] M. O. Cakiroglu, **H. Kurban**†, E. K. Buxton, M. M. Dalkilic. "A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting using TimesNet," *IEEE Access*, 2025.

[28] C. Polat, E. Serpedin, M. Kurban, **H. Kurban**†. "CrysMTM: A Multiphase, Temperature-Resolved, Multimodal Dataset for Crystalline Materials," *Machine Learning: Science and Technology*, 2025.

[27] C. Polat, **H. Kurban**†, M. Kurban. "QuantumShellNet: Ground-State Eigenvalue Prediction of Materials Using Electronic Shell Structures and Fermionic Properties via Convolutions," *Computational Materials Science*, 246, 113366, 2025.

[26] P. Sharma, S. Mishra, **H. Kurban**†, M. M. Dalkilic. "*p*-ClustVal: A Novel *p*-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data," *International Journal of Data Science and Analytics*, 2025.

2024

[25] F. Ay, S. Althunibat, K. Qaraqe, **H. Kurban**†. "A Noise-Adaptive Machine Learning Framework for Optimizing User Grouping in Dynamic IM-OFDMA Systems," *IEEE Transactions on Communications*, 2024.

[24] M. O. Cakiroglu, **H. Kurban**†, K. Qaraqe, L. Aljihmani, G. Petrovski, M. M. Dalkilic. "A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph," *Scientific Reports*, 2024.

[23] C. Polat, M. Kurban, **H. Kurban**†. "Multimodal Neural Network-Based Predictive Modeling of Nanoparticle Properties from Pure Compounds," *Machine Learning: Science and Technology*, 2024.

[22] M. Kurban, C. Polat, E. Serpedin, **H. Kurban**†. "Enhancing the Electronic Properties of TiO_2 Nanoparticles Through Carbon Doping: An Integrated DFTB and Computer Vision Approach," *Computational Materials Science*, 244, 113248, 2024.

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[76] **When Do Errors Compound Across Languages?: Heterogeneous Multilingual Reliability of Long-Horizon Agentic Tool Use.**
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[75] **Knowing When Not to Answer: Evidence-Derived Abstention Requires Evidence That Matches the Claim**
S. Abdaljalil, E. Serpedin, **H. Kurban**†.
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[74] **Gain Ratios Are Not Identifiable: A Corrected Estimator for Paired LLM Evaluation.**
S. Abdaljalil, E. Serpedin, **H. Kurban**†.
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[73] **Ask the Judge Twice: A Label-Free Certificate for Cross-Lingual Reward Hacking.**
A. S. Mahmood, S. Abdaljalil, **H. Kurban**†.
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[72] **When Does Consensus Mean Correctness? Measuring the Agreement–Accuracy Coupling with Semantics-Preserving Re-Rendering.**

R. Khanbayov, **H. Kurban**†.
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[71] **Consistency Has a Computable Blind Spot: A Commutation Theory of Label-Free Reliability for Vision-Language Figure Reading**
R. Khanbayov, **H. Kurban**†.
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[70] **Counterfactual Sensitivity Is Not Repairability: Auditing Replay Probes for Video Evidence**
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[69] **MERIT: Multi-Perspective Evidence-Gated Review for ICU Triage**
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[68] **A Low Grounding Score Is Not an Ungrounded Judge: Identifying the Perceptibility Confound in Multimodal Oversight**
R. Khanbayov, **H. Kurban**†.
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[67] **One Perturbation Is Not Enough: Identifiability and Blind Baselines for Behavioral AI Evaluation**
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[66] **It's the Decoding Format, Not the Perturbation: Auditing Consistency-Based Selection for Vision-Language Test-Time Scaling**
P. Zheng, **H. Kurban**†.
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[65] **Thinking Longer, or Told to Think Harder? Instruction Confounding in Test-Time-Compute Evaluation.**
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[64] **When Does Retrieval Help Time-Series Forecasting**
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[63] **Detect Early, Escalate Rarely: Anytime Detection of AI-Generated Video from the Compressed Bitstream**
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[62] **Conformal Coverage Guarantees for Any Video Temporal Grounder**
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[61] **The Spectrum Is Not Enough: When Context Helps Time-Series Forecasting**
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[60] **4D Synchronized Fields: Motion-Language Gaussian Splatting.**
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[54] **GRAPE: Graph-Augmented Prototype Explanations for Interactive Medical Image Diagnosis.**
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[53] **Perception, Not Reasoning, Limits Visual Theory of Mind.**
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[52] **Rank Reversal in Multilingual LLM Judges: A Label-Free Double-Centering Calibrator.**
A. S. Mahmood, S. Abdaljalil, **H. Kurban**†.
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[51] **Correct but Not Grounded: Stress Testing Evidence Utilization in Large Language Models.**
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[50] **Hydrogen-Rich Surfaces Enhance Storage Capacity in $MgAlH_3$ Perovskite Nanoparticles.**
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[49] **Optimizing EV Charging Infrastructure Utilization via Dynamic Rerouting and Predictive Load.**
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[43] **Perception, Not Reasoning, Limits Visual Theory-of-Mind in Multimodal Foundation Models.**
M. R. Barhdadi, E. Serpedin, **H. Kurban**†.
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[42] **How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science.**
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S. Abdaljalil, K. Qaraqe, E. Serpedin, **H. Kurban**†.
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M. R. Islam, C. Mondal, M. Y. Ansari, **H. Kurban**†, E. Serpedin.
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[27] **PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views.**
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[26] **Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting.**
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[24] **TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials.**
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M. O. Cakiroglu, I. B. Altun, S. R. Fahim, **H. Kurban**†, M. M. Dalkilic, R. Atat, A. Takiddin, E. Serpedin, K. Qaraqe.
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[12] **AReS: An AutoML Regression Service for Data Analytics and Novel Data-centric Visualizations.**
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Open-Source Software and Research Tools

Selected releases. Complete repositories for all projects are linked from the associated papers and from the Kurban Intelligence Lab GitHub organization at github.com/KurbanIntelligenceLab.

- **DCEM**: R package for clustering big data via data-centric modification of Expectation Maximization. Published in *SoftwareX*, 2022.
- **ccImpute**: Consensus-clustering-based imputation of dropout events in single-cell RNA-seq data; R implementation. Published in *BMC Bioinformatics*, 2022; updated version in *SoftwareX*, 2026.
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- **VidAudit**: Six-control protocol and toolkit for auditing generalization in AI-generated video detection. Associated with IEEE/CVF WACV 2027. — UNDER REVIEW
- **STEMGym**: Gymnasium environment for benchmarking dose-efficient autonomous scanning transmission electron microscopy. Associated with NeurIPS 2026. — UNDER REVIEW
- **TDCM25 / CrysMTM / QuantumCanvas**: Multimodal benchmarks for temperature-dependent crystalline materials, materials foundation models, and visual learning of atomic interactions. Associated with ICLR 2025 and MLST 2025.
- **Extended de Bruijn Graph (eDBG)**: Symbolic time-series and biological-sequence representation framework. Associated with *Machine Learning: Science and Technology*, 2024, and *IEEE Access*, 2025.

- **AReS**: AutoML regression service with data-centric visualizations. Associated with ACM KDD Undergraduate Consortium, 2023.

Invited Talks and Presentations

Conference Presentations: Flagship AI/ML Venues

- *Perception, Not Reasoning, Limits Visual Theory of Mind.* 40th NeurIPS, Sydney, Australia, 2026.
- *Rank Reversal in Multilingual LLM Judges: A Label-Free Double-Centering Calibrator.* 31st EMNLP, Budapest, Hungary, 2026.
- *Correct but Not Grounded: Stress Testing Evidence Utilization in Large Language Models.* 31st EMNLP, Budapest, Hungary, 2026.
- *IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video.* 19th ECCV, Malmö, Sweden, 2026.
- *Multilingual Prompt Localization for Agent-as-a-Judge: Language and Backbone Sensitivity in Requirement-Level Evaluation.* 3rd COLM, San Francisco, CA, USA, 2026.
- *Perception, Not Reasoning, Limits Visual Theory-of-Mind in Multimodal Foundation Models.* 43rd IEEE/CVF CVPR, Multi-Modal Reasoning for Agentic Intelligence, Denver, CO, USA, 2026.
- *How Far Can You Grow? Characterizing the Extrapolation Frontier of Graph Generative Models for Materials Science.* 32nd ACM SIGKDD KDD, Jeju, Korea, 2026.
- *Evaluating Trustworthy Multilingual Alignment in LLMs via Synthetic Natural Language Inference.* 35th IJCAI & 29th ECAI, Trust-AI, Bremen, Germany, 2026.
- *EMPATHIA: Multi-Faceted Human–AI Collaboration for Refugee Integration.* 39th Conf. on Neural Information Processing Systems (NeurIPS), San Diego, USA, 2025.
- *SAFE: A Sparse Autoencoder-Based Framework for Robust Query Enrichment and Hallucination Mitigation in LLMs.* 30th Conf. on Empirical Methods in Natural Language Processing (EMNLP) Findings, Suzhou, China, 2025.
- *Theorem-of-Thought: A Multi-Agent Framework for Abductive, Deductive, and Inductive Reasoning in Language Models.* 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Towards Knowledgeable Foundation Models, Vienna, Austria, 2025.
- *Stress-Testing Multimodal Foundation Models for Crystallographic Reasoning.* 63rd ACL, Towards Knowledgeable Foundation Models, Vienna, Austria, 2025.
- *xChemAgents: Agentic AI for Explainable Quantum Chemistry.* 42nd Int. Conf. on Machine Learning (ICML), Multi-Agent Systems in the Era of Foundation Models, Vancouver, Canada, 2025.
- *Multivariate de Bruijn Graphs: A Symbolic Graph Framework for Time Series Forecasting.* 42nd ICML, Foundation Models for Structured Data (FMSD), Vancouver, Canada, 2025.
- *PhysicsNeRF: Physics-Guided 3D Reconstruction from Sparse Views.* 42nd ICML, Building Physically Plausible World Models, Vancouver, Canada, 2025.
- *Beyond Atomic Geometry Representations in Materials Science: A Human-in-the-Loop Multimodal Framework.* 42nd ICML, DataWorld, Vancouver, Canada, 2025.

- *Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding.* 42nd IEEE/CVF CVPR, MM4Mat, Nashville, TN, USA, 2025. SPOTLIGHT
- *TDCM25: A Multi-Modal Multi-Task Benchmark for Temperature-Dependent Crystalline Materials.* 13th Int. Conf. on Learning Representations (ICLR), Singapore, 2025.
- *Learning Physics Like Humans: Uncovering Physical Laws from Monocular Videos.* 20th International Conf. on Computer Vision (ICCV, Human-inspired Computer Vision), Hawaii, USA, 2025.
- *Exploring Various Sequential Learning Methods for Deformation History Modeling.* 26th Engineering Applications of Neural Networks / Engineering Applications and Advances of Artificial Intelligence (EAAAI), Limassol, Cyprus, 2025.
- *What Data-Centric AI Can Do For k-means: A Faster, Robust k-means-d.* 41st ICML, Data-centric Machine Learning Research (DMLR), Vienna, Austria, 2024.
- *p-ClustVal: A Novel p-adic Approach for Enhanced Clustering and Valuation in High-Dimensional scRNASeq Data.* 11th IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA), San Diego, USA, 2024.
- *A Novel Discrete Time Series Representation with De Bruijn Graphs for Enhanced Forecasting using TimesNet.* (Extended Abstract) 11th IEEE International Conf. on Data Science and Advanced Analytics (DSAA), San Diego, USA, 2024.
- *Novel NBA Fantasy League Driven by Engineered Team Chemistry and Scaled Position Statistics.* IEEE Int. Conf. on Big Data, Data-Centric AI, Sorrento, Italy, 2023.
- *Are Sports Awards About Sports? Using AI to Find the Answer.* ECML/PKDD 10th Workshop on Machine Learning and Data Mining for Sports Analytics, Turin, Italy, 2023.
- *AReS: An AutoML Regression Service for Data Analytics and Novel Data-centric Visualizations.* 29th ACM SIGKDD International Conf. on Knowledge Discovery and Data Mining, Undergraduate Consortium (KDD-UC), Long Beach, CA, USA, 2023.
- *Are They What They Claim: A Comprehensive Study of Ordinary Linear Regression Among the Top Machine Learning Libraries in Python.* 29th ACM SIGKDD KDD-UC, Long Beach, CA, USA, 2023.
- *Data Expressiveness and Its Use in Data-centric AI.* NeurIPS Data-centric AI Workshop, 2021.
- *Using Data Analytics to Optimize Public Transportation on a College Campus.* IEEE DSAA, Turin, Italy, 2018.
- *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure.* IEEE Int. Conf. on Big Data, Boston, MA, USA, 2017.
- *Improving Expectation Maximization Algorithm over Stellar Data.* Workshop on Management, Search and Mining of Massive Repositories of Solar and Stellar Astronomy Data, Boston, MA, USA, 2017.
- *Case Study: Clustering Big Stellar Data with EM*.* IEEE/ACM BDCAT, Austin, TX, USA, 2017.
- *EM*: An EM Algorithm for Big Data.* IEEE DSAA, Montreal, Canada, 2016.
- *EM*: An EM Algorithm for Big Data.* HONORABLE MENTION PAPER AWARD. IEEE DSAA, Montreal, Canada, 2016.

- *A New Set of Random Forests with Varying Dynamic Data Reduction and Voting Techniques.* IEEE DSAA, Shanghai, China, 2014.

Conference Presentations: Domain and Applied Venues

- *Hydrogen-Rich Surfaces Enhance Storage Capacity in $MgAlH_3$ Perovskite Nanoparticles.* 52nd IECON, Doha, Qatar, 2026.
- *Optimizing EV Charging Infrastructure Utilization via Dynamic Rerouting and Predictive Load.* 52nd IECON, Doha, Qatar, 2026.
- *Edge-Conditioned Contrastive Graph Learning for False Data Injection Detection in Transmission Power Systems.* 52nd IECON, Doha, Qatar, 2026.
- *Physics-Informed Data Augmentation and Explainable Machine Learning for Hydrogen Storage Prediction in $LiNH_6$ Nanoparticles.* 52nd IECON, Doha, Qatar, 2026.
- *Machine Learning-Based Robust Distributed Economic Dispatch Under Data Integrity Attacks.* 17th IEEE SmartGridComm, College Station, TX, United States, 2026.
- *Monte Carlo Dropout for Robust and Uncertainty-Aware Lung Nodule Classification.* 34th EUSIPCO, Bruges, Belgium, 2026.
- *Fair-FL: Verifiable Inclusive Aggregation for Fair Federated Learning.* 34th EUSIPCO, Bruges, Belgium, 2026.
- *Strategic Bidding in Competitive and Adversarial Electricity Markets Using No-Regret Learning Algorithms.* 23rd IFAC World Congress, Busan, Korea, 2026.
- *Site Selectivity Controls Reversible Hydrogen Adsorption and Release on TiFe Nanoparticles: An Interpretable Machine Learning-Assisted Study.* 6th International Conf. on Electrical, Computer and Energy Technologies (ICECET), Rome, Italy, 2026.
- *Agentic and Explainable Intelligent Screening of Metal–Organic Frameworks for CO_2 Direct Air Capture.* 8th Intelligent and Fuzzy Systems Conf. (INFUS), Ankara, Turkey, 2026.
- *Thermodynamic and Electronic Mapping for Hydrogen Storage on Zr-Decorated TiO_2 Nanoparticles with an Interpretable Learning Agent and Computational Modeling.* 25th International Conf. on Advanced Nanomaterials (ANM), Aveiro, Portugal, 2026.
- *De Bruijn Graph-Enhanced Time Series Models for Electricity Load Forecasting.* 17th International Symposium on Signals, Circuits and Systems (SSCS), Iaşi, Romania, 2025.
- *Data-Efficient Hydrogen Adsorption Prediction in Copper Nanoclusters: A Computer Vision-Based Transfer Learning Approach.* 19th International Conf. on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Turkey, 2025.
- *Instance-Based Learning-Driven Density of States Analysis in Functionalized Fullerene Derivatives for Optimizing Organic Photovoltaics.* 19th CPE-POWERENG, Antalya, Turkey, 2025.
- *Decentralized N-1 Contingency Analysis for Cascading Failure Prediction in Multi-Region Power Systems using Consortium Blockchain.* 5th International Conf. on Electrical, Computer and Energy Technologies (ICECET), Paris, France, 2025.
- *Predicting Optical Bandgaps in C_{60} and Functionalized Derivatives from Limited Data for Renewable Energy Applications.* 19th CPE-POWERENG, Antalya, Turkey, 2025.
- *Size Dependent Electronic Structure and Structural Properties of Cupric Oxide (CuO) Nanoparticles.* International Conf. on Nanomaterials and Energy (NEM), 2019.

- *Employing Software Engineering Principles to Enhance Management of Climatological Datasets for Coral Reef Analysis.* 6th International Workshop on Climate Informatics, Colorado, USA, 2016.
- *Red-RF: Reduced Random Forest for Big Data using Priority Voting & Dynamic Data Reduction.* IEEE BigData Congress, New York, USA, 2015.

Invited Academic Colloquia and Lectures

- *Data-Centric Machine Learning.* Invited colloquia at San Jose State University, University of Houston, University of Montana, American University of the Middle East, United Arab Emirates University, University of Illinois, Zayed University, University of Missouri, and University of South Carolina, 2022 to 2023.
- *Clustering in Data Science.* Department of Statistics and Data Science, Northwestern University, 2022.
- *Comparison of Machine Learning Algorithms for CuO Nanoparticles.* 4th Int. Conf. on Physical Chemistry and Functional Materials (PCFM21), Elazığ, Turkey, 2021.
- *Practical Data Science: Examining the Correlations between Structural and Electronic Properties of Different Phases of TiO_2 Nanoparticles.* Int. Conf. on Advanced Technologies (ICAT), Istanbul, Turkey, 2020.
- *Predicting Atom Types of Anatase TiO_2 Nanoparticles with Machine Learning.* Int. Conf. on Engineering and Innovative Materials (ICEIM), Singapore, 2020.
- *Introduction to Data Science.* Computer Engineering Department, Izmir Institute of Technology, 2020.
- *Clustering Techniques in Engineering.* Computer Engineering Department, Yildirim Beyazit University, 2019.
- *Data Science Fundamentals.* Informatics Department, Istanbul Technical University, 2018.
- *EM Algorithms for Clustering.* Computer Science Department, Indiana University, 2016.
- *Principal Component Analysis.* Computer Science Department, Indiana University, 2016.
- *Data Structures and Algorithms.* Computer Science Department, Indiana University, 2014.
- *Ensemble Models in Machine Learning.* Computer Science Department, Indiana University, 2013.

Industry, Medical, and Leadership Forums

- *AI-Driven Innovation in Medicine: Success, Failure & Challenges.* 3rd Surgical Annual Research Day, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients.* AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.
- *BlinkAI: A Non-Verbal Communication Tool for Locked-In Syndrome Patients.* Web Summit, Doha, Qatar, 2025.
- *A Reinforcement Learning Approach to Effective Forecasting of Pediatric Hypoglycemia in Diabetes I Patients: An Extended de Bruijn Graph.* AI and Medicine Symposium, Sidra Medicine, Doha, Qatar, 2025.
- *AI-Powered Innovation: Driving Organization.* Future Leaders Programme, Muscat, Oman, 2024.
- *Data Science and Big Data.* Eli Lilly and Company, 2017.

TEACHING EXPERIENCE

Instructor of Record

2024 TO PRESENT

- **ICT 706:** Special Topics in Vision-Language Models (Graduate), Spring 2026.

2023 TO PRESENT

- **ELEN & ECEN 250:** Machine Learning for Electrical Engineering (Undergraduate), Spring 2026.
- **ECEN 491:** Independent Study: Research (Undergraduate), Spring 2025, Summer 2025, Fall 2025, Spring 2026.
- **ECEN 485:** Independent Study: Directed Studies (Undergraduate), Spring 2026.
- **ENGR 110:** Introduction to Programming (Undergraduate), Fall 2024, Spring 2025, Fall 2025, Fall 2026.
- **ENGR 102:** Engineering Lab I: Computation (Undergraduate), Spring 2024, Fall 2024, Spring 2025.
- **ECEN 210:** Computer Programming and Algorithms (Undergraduate), Fall 2023.
- **ECEN 248:** Introduction to Digital Systems Design (Undergraduate), Fall 2023.

2017 TO 2023

- **CSCI-B 565:** Data Mining (Graduate), Spring 2023.
- **CSCI-B 365:** Introduction to Computers and Programming (Undergraduate), Spring 2023.
- **CSCI-C 241:** Discrete Structures for Computer Science (Undergraduate), Summer 2022, Fall 2022.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2022.
- **CSCI-B 505:** Applied Algorithms (Graduate), Fall 2021.
- **CSCI-P 556:** Applied Machine Learning (Graduate), Fall 2017.
- **INFO-I 590:** Online Applied Data Mining (Graduate), Fall 2017.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2018.
- **CSCI-B 365:** Introduction to Data Analysis and Mining (Undergraduate), Spring 2018.

2018 TO 2021

- **BMH 101:** Algorithms and Programming I (Undergraduate), Spring 2019, Fall 2019.
- **BMH 406:** Data Security (Undergraduate), Spring 2019.
- **BMH 104:** Web and Internet Technologies (Undergraduate), Spring 2019.
- **BMH 205:** Data Structures (Undergraduate), Fall 2018.
- **BMH 413:** Artificial Neural Networks (Undergraduate), Fall 2018, Fall 2019.
- **BMH 103:** Introduction to Computer Engineering (Undergraduate), Fall 2018, Fall 2019.

Associate Instructor

- **CSCI-B 565:** Data Mining (Graduate), Fall 2012, Fall 2013, Spring 2015, Spring 2016, Fall 2016.
- **CSCI-B 351:** Elements of Artificial Intelligence (Undergraduate/Graduate), Spring 2017.
- **CSCI-B 555:** Machine Learning (Graduate), Spring 2013.
- **CSCI-C 343:** Data Structures (Undergraduate/Graduate), Spring 2014.
- **CSCI-B 365:** Seminar in Computer Science: Data Mining (Undergraduate), Fall 2014, Fall 2015.
- **CSCI-B 609:** Topics in Algorithms and Computing Theory (Graduate), Fall 2014.
- **INFO-I 590:** Real World Data Science (Graduate, Online), Summer 2016, sponsored by Eli Lilly.

Research Supervision and Mentoring

Summary: 1 postdoctoral researcher in post, 1 completed • 6 Ph.D. researchers in training, 1 graduated • 4 M.Sc. researchers in training, 3 graduated • 3 undergraduate researchers, 2 graduated • 23 capstone students across 6 teams, all graduated.

Postdoctoral Researchers

Current

- **Dr. Jian Jin** (postdoctoral researcher). Research areas: generative AI; diffusion and score-based generative models; representation learning for multimodal and scientific data.

Graduated

- **Dr. Mehmet Tuncel** (postdoctoral researcher, graduated). *Current position:* Istanbul Technical University (İTÜ), Türkiye.
  Research areas: Physics-informed neural networks and curriculum-based adaptive sampling for stiff PDEs; agentic and explainable AI for quantum chemistry; multimodal property prediction for functionalized fullerenes; 3D and 4D Gaussian scene representations; human–AI collaboration systems.
  Selected outcomes: NeurIPS 2025 • ICML 2025 • 3DV 2027.

Doctoral Researchers

Current

- **Can Polat** (Ph.D. candidate). Equivariant and symmetry-aware interatomic potentials (Clifford algebra, parity constraints); scientific foundation models and AI for science; grounding, hallucination, and consistency auditing of materials foundation models; multimodal benchmark construction for crystalline materials and quantum chemistry; agentic AI for scientific reasoning; active-learning-assisted hydrogen-storage screening.
  Selected outcomes: *npj Computational Materials* • KDD 2026 • ICML 2025 • ACL 2025 • ICLR 2025 • CVPR 2025 (Spotlight).
- **Mert Onur Cakiroglu** (Ph.D. candidate). Symbolic graph representations for temporal data (de Bruijn and multivariate de Bruijn graphs); long-horizon and retrieval-augmented time-series forecasting; audit protocols for forecasting claims against strong baselines; AI-generated video detection and cross-generator generalization; learned geometric quantization for image tokenization.

Selected outcomes: ICML 2025 • *Scientific Reports* • *Machine Learning: Science and Technology* • AAAI 2027 • WACV 2027.

- **Samir Abdaljalil** (Ph.D. candidate). Hallucination detection and mitigation in large language models (SINdex, HalluVerse3, SAFE); sparse-autoencoder feature analysis for LLM safety; evidence-grounded reasoning and abstention; identifiability and corrected estimators for paired LLM evaluation; multi-agent reasoning frameworks (Theorem-of-Thought); calibration-aware clinical triage.

  Selected outcomes: EMNLP 2025 and 2026 • ACL 2025 • *Artificial Intelligence* • AAAI 2027 • 18 co-authored papers.

- **Md Rabiul Islam** (Ph.D. candidate). Interpretable multimodal medical AI; pulmonary-nodule diagnosis from CT with radiology- and anatomy-aware text fusion; Bayesian decision fusion under diameter priors; uncertainty quantification and calibration (Monte Carlo dropout, selective deferral); agentic AI for biomedical and clinical workflows.

  Selected outcomes: *IEEE Transactions on Artificial Intelligence* • *IEEE Journal of Biomedical and Health Informatics* • *Artificial Intelligence Review* • EUSIPCO 2026.

- **Idil Bilge Altun** (Ph.D. student). Large-scale scientometric mapping of the AI safety literature; hybrid symbolic–neural representations for sequential data; benchmarking for industrial condition monitoring (dissolved gas analysis, long-term load forecasting); learned geometric quantization for image tokenization; multimodal and video foundation models.

  Selected outcomes: *Journal of Artificial Intelligence Research* • ICML 2025 • AAAI 2027 • WACV 2027 • *Engineering Applications of Artificial Intelligence*.

- **Rasul Khanbayov** (Ph.D. student). Label-free reliability and identifiability theory for multimodal evaluation; counterfactual auditing of vision–language judges and overseers; conformal coverage guarantees for video temporal grounding; inverse identification of physical dynamical systems from monocular video; prototype-based explanation for medical imaging.

  Selected outcomes: NeurIPS 2026 • ECCV 2026 • *npj Computational Materials* • AAAI 2027 • EACL 2027.

Graduated

- **Dr. Parichit Sharma** (Ph.D., graduated). *Current position:* University of Maryland, Baltimore, MD, USA.

  Dissertation: Data-Centric Principles for Machine Learning: From Efficient Clustering to Single-Cell Discovery.

  Research areas: Data-centric machine learning; bound-free and energy-efficient clustering (Geometric-k-means, k-means-d); p-adic valuation for high-dimensional data; dropout imputation and clustering for single-cell RNA-seq; reproducible research tooling (DCEM, ccImpute); instance-based learning for materials property prediction.

  Selected outcomes: Machine Learning • *Scientific Reports* • ICML 2024 • 16 co-authored papers.

Master's Students

Current

- **Alhasan Mahmood** (M.Sc. student). Research areas: multilingual LLM-as-a-judge evaluation; rank reversal and label-free double-centering calibration; cross-lingual reward-hacking certificates; agent-as-a-judge prompt localization; multilingual reliability of long-horizon agentic tool use.
  Selected outcomes: COLM 2026 • EMNLP 2026 • EACL 2027.
- **Mohamed Rafi Dheen Meeran** (M.Sc. student). Research areas: requirement-level evaluation of LLM agents; judge robustness under prompt and backbone variation; benchmark design for agentic systems.
- **Puzhuo Zheng** (M.Sc. student). Research areas: test-time scaling for vision–language models; decoding-format confounds in consistency-based selection; auditing protocols for multimodal inference-time compute; representation learning for vision.
  Selected outcomes: AAAI 2027.
- **Marcus Wein Monteiro** (M.Sc. student). Research areas: efficient test-time reasoning for small multimodal models; compute-constrained inference.

Graduated

- **Ganesh Arkanath** (M.Sc., graduated). *Current position:* Meta. *Thesis: Advancing NBA Playoffs Prediction and Fantasy Leagues Using Feature Engineering, Deep Learning, and Explainable AI.* Research areas: sports analytics; engineered team-chemistry and scaled-position features; explainable deep learning for outcome prediction.
- **Madhavan Kalkunte Ramachandra** (M.Sc., graduated). *Current position:* ServiceNow. Research areas: high-dimensional similarity search; *k*-nearest-neighbor acceleration and telescope indexing; AutoML regression services; empirical auditing of production machine learning libraries; scalable ML systems.
- **Jaydeep Chauhan** (M.Sc., graduated). *Current position:* Eli Lilly and Company. *Thesis: Out-of-Distribution Generalization, Spurious Correlations, and Supervised Contrastive Groupwise Learning.* Research areas: out-of-distribution generalization; shortcut and spurious-correlation mitigation; groupwise contrastive learning.

Undergraduate Researchers

Current

- **Mohamed Rayan Barhdadi.** *Thesis: The Physical World as the First Teacher: Structured 4D Scene Representations.* Research areas: physics-guided 3D and 4D reconstruction (PhysicsNeRF, Gaussian splatting); visual theory of mind in multimodal models; inverse identification of physical dynamics from monocular video; human–AI collaboration systems.
  Selected outcomes: NeurIPS 2025 and 2026 • ECCV 2026 • ICML 2025 • ICCV 2025 • AAAI 2027 • 3DV 2027.
- **Rama Alhamidi.** Research areas: agentic video understanding and question answering; counterfactual sensitivity and repairability of replay probes in video agents; cost-aware audit protocols for dynamic tool synthesis; robotics and embodied perception.
  Selected outcomes: EACL 2027 • WACV 2027.
- **Aseel Mohamed.** Research areas: conformal prediction and distribution-free coverage guarantees for video temporal grounding; evaluation protocols for agentic video question answering; robotics and embodied perception.
  Selected outcomes: AAAI 2027 • WACV 2027.

Graduated

- **Sam Johnson** (graduated). Projects: AutoML regression benchmarking (AReS); empirical analysis of regression algorithms; data-centric diagnostics.
  Selected outcomes: KDD Undergraduate Consortium 2023 (two papers).
- **Josh Elms** (graduated). Projects: AutoML regression platform; visual analytics for model interpretability; large-scale regression behavior analysis.
  Selected outcomes: KDD Undergraduate Consortium 2023 (two papers).

Capstone and Senior Design Teams

6 teams, 23 students; all graduated.

- **Ahmed Al-Thani, Khalifa Al-Mansoori, Mohammed Baghazal, Ziyad Ghalayini.** Project: A Novel Extended de Bruijn Graph Framework for Hypoglycemia Prediction. • Best Capstone Project Award
- **Abdulaziz AlBalushi, Mohammed AlRasheed, Nasser AlHababi, Mohammed AlNaimi.** Project: Privacy-Preserving Video-Based Emotion Recognition.
- **Fadwa Abbas, Noor Alabudi, Roudha Al-Khaldi, Sandra Mansour.** Thesis: A Robotics-Assisted Validation Framework for Wi-Fi-Based Indoor Localization. • Best Capstone Project Award
- **Iqra Yakub, Bashar Farooq, Ali Al-Majid, Abdellatif Hussine.** Thesis: A Privacy-Preserving Framework for Elderly Activity Monitoring Using Pose Estimation. • Multiple Best Capstone Awards
- **Noor Ahmed Al-Thani, Muneera Al-Qahtani, Aljazi Naqadan, Al-reem Al-Baker.** Thesis: An AI-Based Monitoring System for Long-Term Load Forecasting.
- **AlMaha AlEmadi, Aisha Al-Kuwari, Shahd Al-Kubaisi.** Thesis: Design and Implementation of a Web-Based Clinical Decision Support Service for Multimodal Lung Cancer Detection.

Professional Service and Leadership

Editorial and Advisory Roles

2023 to present	Editorial Board Member, *Scientific Reports*
2023 to present	Technical Editor, Manning Publications
2023 to 2024	Data Science Advisory Board Member, Pearson Education

Conference Organization and Program Committees

- **Shared Task Organizer**: "Ayn 2026: Cultural Grounding in Arabic Multimodal Generation and Understanding," ArabicNLP 2026 Shared Tasks Program, co-located with EMNLP 2026, Budapest, Hungary, October 24–29, 2026.
- **Technical Program Committee Member**: IEEE International Conference on Blockchain (IEEE Blockchain), 2026.
- **Technical Program Committee Member**: IEEE Cybermatics Congress, Montbéliard, France, 2026.
- **Lead Organizer**: "AI for Science Symposium" (HBKU Horizon Hub, 1st Cycle), Doha, Qatar, January 21–22, 2026.

- **Program Committee Member**: Industry Track, The Web Conference (WWW), Dubai, UAE, 2026.
- **Invited Speaker**: Workshop on "The Role of AI in Strengthening the Resilience of Future Power Systems," Doha, Qatar, 2025.
- **Session Chair**: 4th Digital Flow Assurance Symposium (HBKU, SLB, Wood, TAMUQ), Doha, Qatar, 2025.
- **Organizing Committee**: 19th CPE-POWERENG, Special Session on AI-Driven Innovations in Renewable Energy, 2025.
- **Organizing Committee**: 11th IEEE DSAA, Special Session on Advancing Materials Science through Data Science, 2024.
- **Program Committee Member**: Basarim High-Performance Computing Conference (2020, 2022, 2024).
- **Session Chair**: IJCAI, Jeju, South Korea, 2024.
- **Session Chair**: 11th IEEE DSAA, San Diego, USA, 2024.
- **Session Chair**: IEEE International Conference on Big Data, Sorrento, Italy, 2023.

Peer Review

Conferences

NeurIPS • ICML • ICLR • AAAI • ACM SIGKDD • AISTATS • COLM • IECON • Basarim High-Performance Computing Conference.

Named a *Silver Reviewer* at ICML 2026, a top-reviewer designation based on area-chair ratings of review quality.

Journals

IEEE Transactions on Artificial Intelligence • IEEE Transactions on Neural Networks and Learning Systems • IEEE Transactions on Circuits and Systems I: Regular Papers • Machine Learning: Science and Technology • Frontiers in Artificial Intelligence • Big Data Mining and Analytics • Journal of Chemical Information and Modeling • Information Sciences • Expert Systems with Applications • Engineering Applications of Artificial Intelligence • Applied Artificial Intelligence • Journal of Medical Internet Research • Computers in Biology and Medicine • Journal of Chemical Information and Modeling • Materials & Design • Applied Energy • Energy Conversion and Management • Journal of Energy Storage • eTransportation • Electric Power Systems Research • International Journal of Electrical Power and Energy Systems • Digital Communications and Networks • Concurrency and Computation: Practice and Experience • Systems Science & Control Engineering • Results in Engineering • Behaviour & Information Technology • International Journal of Human-Computer Interaction • Frontiers in Psychiatry • IEEE Access • Entertainment Computing • Transactions on Consumer Electronics • Machine Intelligence Research • CAAI Transactions on Intelligence Technology • Frontiers of Computer Science • Optical and Quantum Electronics • Frontiers in Big Data • SN Computer Science • Sustainable Energy, Grids and Networks • ACM Transactions on Intelligent Systems and Technology • Pediatric Diabetes • Physica B: Condensed Matter.

University Service

- **Hamad Bin Khalifa University**, Division of Computing and Electrical Engineering (2026 to 2027): *Co-Chair (PhD)*, Postgraduate Admission Committee; Curriculum and Quality Assurance Committee; Student Recruitment, Development and Activities Committee.

- **Hamad Bin Khalifa University** (Fall 2024 to Spring 2026): Industrial & Government Outreach Committee; ABET & Curriculum; Academic Advising; AI for Science Initiative.
- **Texas A&M University at Qatar** (Spring 2024): ABET & Curriculum; Seminars & Invited Speakers; Budget.
- **Texas A&M University at Qatar** (Fall 2023): ABET & Curriculum; Seminars & Invited Speakers; Budget; Teaching Load Model.

Academic and Professional Appointments

2024 to present **Assistant Professor**, College of Science and Engineering
Hamad Bin Khalifa University, Doha, Qatar

2023 to present **Adjunct Associate Professor**, Computer Science Dept. & Data Science Program
Indiana University Bloomington, IN, USA

2024 to present **Adjunct Assistant Professor**, Electrical & Computer Engineering
Texas A&M University at Qatar, Doha, Qatar

2023 to 2024 **Assistant Professor**, Electrical & Computer Engineering
Texas A&M University at Qatar, Doha, Qatar

2021 to 2023 **Visiting Associate Professor**, Computer Science Department
Indiana University Bloomington, IN, USA

2019 to 2021 **Assistant Professor**, Computer Engineering Department
Siirt University, Siirt, Türkiye

2017 to 2019 **Visiting Assistant Professor**, Computer Science Department
Indiana University Bloomington, IN, USA

2015 to 2017 **Undergraduate Research Mentor**, Computer Science Department
Indiana University Bloomington, IN, USA

2015 to 2016 **Data Scientist**, Turbo Appeal, Chicago, IL, USA

Education

2017 **Ph.D. in Computer Science**, Minor in Statistics
Indiana University Bloomington, IN, USA
Dissertation: *A Novel Approach to Optimization of Iterative Machine Learning Algorithms: Over Heap Structure*
Advisor: Prof. Mehmet M. Dalkilic
Committee: Prof. Predrag Radivojac • Prof. Michael W. Trosset • Prof. Yuzhen Ye

2012 **MSc in Computer Science**
Indiana University Bloomington, IN, USA

2010 **Certificate in Business Communication**
University of Connecticut, Storrs, CT, USA
Non-degree professional certificate program

2008 **BSc in Mathematics**
İnönü University, Malatya, Türkiye

Honors and Awards

2026 SILVER REVIEWER: 43rd Int. Conf. on Machine Learning (ICML), awarded on area-chair ratings of submitted reviews

2026 EDITORIAL CONTRIBUTION AWARD: Springer Nature "Editor of Distinction Awards," for editorial service to *Scientific Reports*

2025 CVPR SPOTLIGHT: Multimodal Learning for Materials Science (MM4Mat), Nashville, TN, USA

2017 BEST POSTER AWARD: IEEE/ACM Int. Conf. on Big Data Computing, Applications & Technologies (BDCAT), Austin, TX

2016 HONORABLE MENTION PAPER AWARD: IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA), Montreal, Canada

2016 to 2017 Nomination for Researcher of the Year, Indiana University Bloomington

2014 to 2015 Nomination for Associate Instructor of the Year, Indiana University Bloomington

2010 to 2012 Computer Science Graduate Fellowship, Indiana University Bloomington

2009 to 2017 Turkish National Ministry of Education Scholarship (comprehensive graduate studies funding)

Professional Memberships

Institute of Electrical and Electronics Engineers (IEEE) • Association for Computing Machinery (ACM).

Technical Skills and Languages

Machine learning: PyTorch • JAX • TensorFlow • Hugging Face Transformers • scikit-learn • PyTorch Geometric • vLLM.

LLMs and agents: LangChain • LangGraph • LlamaIndex • OpenAI and Anthropic APIs • RAG pipelines • LoRA/QLoRA fine-tuning • LLM evaluation and red-teaming.

Systems and HPC: Python, C/C++, R, MATLAB • SLURM • distributed training (DDP, FSDP) • CUDA • Docker, Git, Linux • Weights & Biases, MLflow.

Languages: English (fluent, academic and professional) • Turkish (native).

References

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